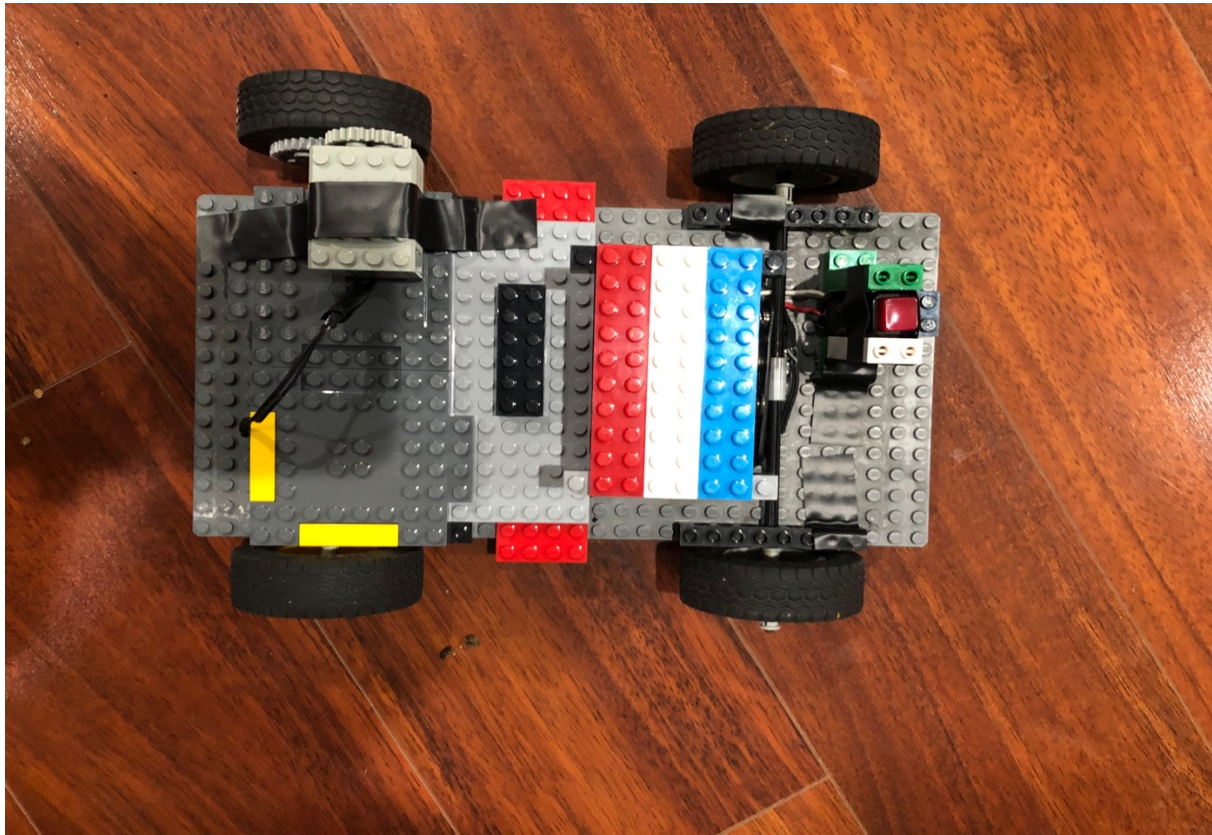


SRVUSD Science Olympiad 2020

Battery Buggy



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Metamorphic Mastermind

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Introduction

An electric car is driven by a DC electric motor, powered by two or four batteries. Transmission of force from the motor to the car axle is accomplished using two gears. A simple on and off switch is used to connect or disconnect the circuit. Large Lego wheels along with rubber rings are used as tires. The car chassis is made of a Lego baseboard, reinforced by Lego blocks that will also support the axles of the gears of the wheels. The Legos are also used to build the battery and switch base of the buggy.

Alternative Design

We also tried an alternative model using wood (details indicated in Appendix A) but that didn't work out as good as the Lego- based battery buggy detailed here.

Statement of Purpose

1. To understand how to construct a vehicle using electrical components: batteries, wires, motors, and other components like axles and gears.
2. To understand how electrical energy can move a vehicle.
3. To understand how the design affects its ability move in a straight line.
4. To understand how the design affects its ability move faster.

Materials

- * Lego Bricks
- * Lego Base
- * Lego battery holder (built using Lego parts)
- * Lego axles
- * Lego connectors
- * Lego wheels (2.5 in, 2in)
- * Lego motor
- * Circuit wires
- * 4 AA batteries
- * Lego Gears
- * Lego Switch Holder (built using Lego parts)

Design

Kinetic Energy

- 1.Experiment: Change Mass
- 2.Conclusion: Faster speed.
- 3.Experiment: Motor Strength
- 4.Conclusion: Greater voltage. Faster speed.

Body

- 1.Connect all of the other parts/systems together
- 2.Easy to design and build
- 3.Broader wheelbase makes it travel longer distance in straight line

Wheels and Axles

- 1.Large diameter Wheel (2.75 – Speed is 4.25 sec/ft)
- 2.Revolves fewer times to travel a given distance and is faster
- 3.Small diameter wheel (2.25 – Speed is 4 sec/ft)
- 4.Revolves more times to travel a given distance and is slower.
- 5.The axle stoppers support and hold the axles tightly together.

Electrical Systems

1.5V, 4 batteries & Wires

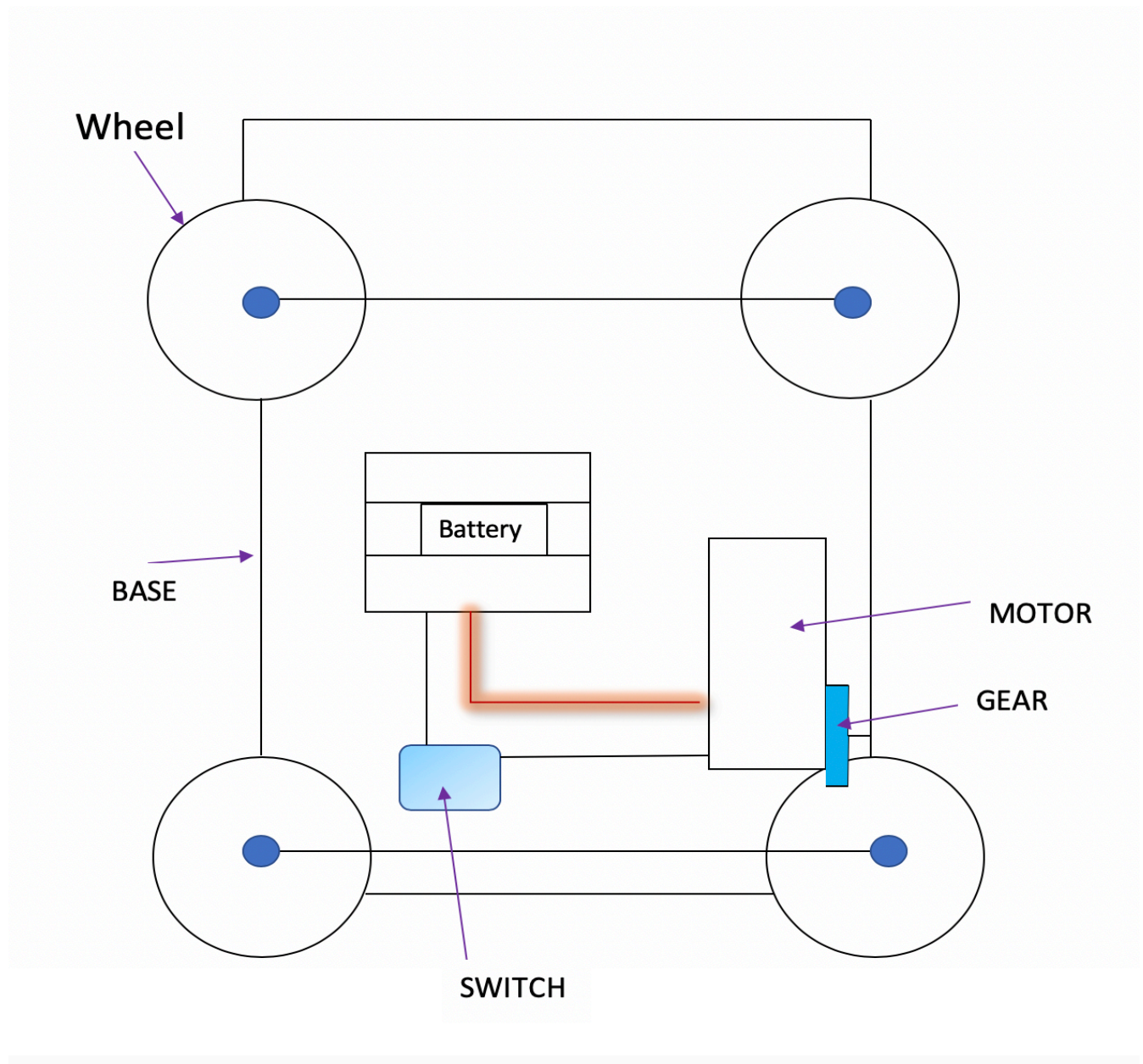
Procedure

1. Build a base from Legos and including some bordering bricks for the wheels. Place the axles through a Lego brick. Put the gear on the back-left side (connected to the motor) and on the wheel
2. Make a battery holder for the battery. Place the wires together (negative and positive) attach with the motor wires.
3. Make a holder for the switch. Connect the switch with the wires.
4. Measure 25 feet and time how long it takes for the buggy to cover that distance. Divide this number by 25 to get the speed of the buggy.

Wheel Diameter	Speed
2.5 inches	0.3sec/feet
2 inches	0.43sec/feet

5. Repeat for wheels with different diameters.

Circuit Diagram



Lessons Learnt

How does wheel diameter affect the speed of a vehicle?

Conclusion: If we increase the wheel diameter, then the speed will increase. This happens because the wheel can cover more distance in one rotation.

How does motor strength affect the speed of a vehicle?

Conclusion: If we increase the strength of the motor, the speed of the buggy increases.

How does the buggy travel in a straight line?

Conclusion: If the axles are aligned with the wheels and are fixed tightly, broader wheelbase, the buggy travels in the straight line.

Project Conclusion

We have finally finished building our battery car. We learned that by connecting the gear ratio correctly using a good strength motor, making wheels of larger diameter our vehicle can travel faster. By using broader wheelbase, parallel axles and proper alignment helps to move in straight line. Building it using Legos, is easier, flexible and lighter and also allows us to customize it in many ways.



Buggy Photos







Appendix A (Alternate Model)

An alternative battery buggy model using wood

Hypothesis

- A battery buggy made with wood may be faster and travel in straight line.

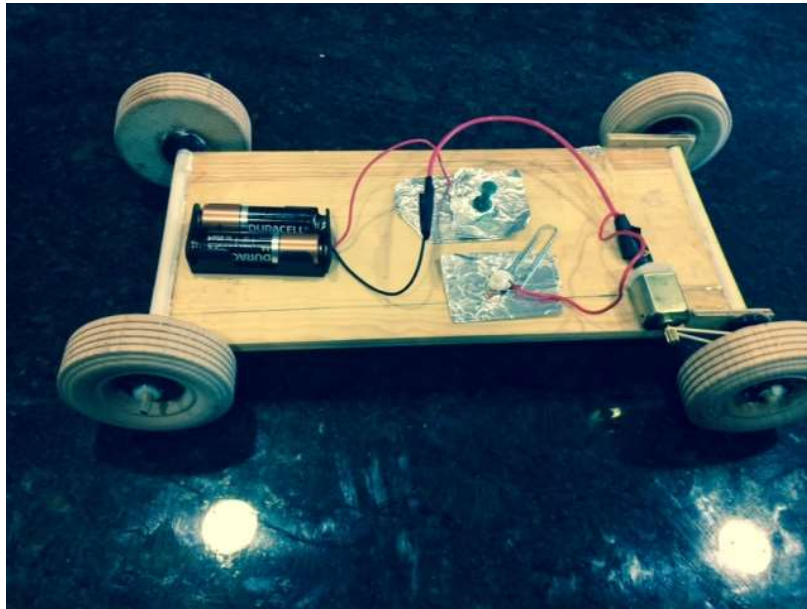
Materials

- Wooden plank (cut a small sized from Home Depot)
- Battery holder
- Wooden Axles
- Rubber band
- Switch (using paperclips, aluminum foil and thumbnails)
- wheels (2.5in, 2in)
- Motor
- Circuit wires
- 2 AA batteries

Procedure

- Build a base using a wooden plank. Attach wooden axle. pulleys on the back-left side (connected to the motor) and on the wheel using rubber bands to enable wheel rotation.
- Use a battery holder for the battery. Place the wires together (negative and positive) attach with the motor wires.

- Build a switch using foil, paper clips to switch on and off. Connect the switch with the wires.
- Measure 25 feet and time how long it takes for the buggy to cover that distance. Divide this number by 25 to get the speed of the buggy.



Conclusion

The experiment was not successful because the buggy could not travel in a straight line. We tried to add pulleys and align the wheels properly, used a wider base wheel, but the buggy did not travel straight. We had to use Legos instead and try the above procedure. The experiment was now successful. The wooden car didn't move straight because we could not get the wheels to align properly.